

Felipe Espinosa

AI Research Engineer

Paris, France | +33 7 66 09 03 35 | felipeespinosa.13@outlook.com

Portfolio | LinkedIn | GitHub | Google Scholar

Education

Université Paris-Saclay, France

Sep. 2023–Sep. 2025

M.Sc. in Computer Science, Artificial Intelligence

Average: 16.2/20; Mention Très Bien

- Selected coursework: Reinforcement Learning, Computer Vision, Signal Processing, Advanced Optimization, Deep Learning, Probabilistic Generative Models, Learning Theory and Advanced Machine Learning, and Image Mining.

Lloyds International College, Sydney, Australia

Feb. 2020–Apr. 2022

Advanced Diploma of Leadership and Management (BSB61015)

Universidad Industrial de Santander, Colombia

Feb. 2010–Dec. 2016

B.Sc. in Electronic Engineering

GPA: 3.89/5.00

- Selected coursework: Digital Control Systems, Signal Processing, Discrete Signal Processing, C/C++ Programming, Computer Architecture, Numerical Computation, and Probability and Statistics.
- Academic exchange at the University of São Paulo, Brazil (Mar.–Nov. 2014); selected coursework: Control Systems, Process Automation, Intelligent Systems, and Electronic Systems Design.

Research Experience & Selected Projects

Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), Orsay, France

Feb. 2024–Aug. 2024

AI Research Intern, Speech Synthesis

Supervisors: Philippe Boula de Mareüil and Marc Evrard

- Developed the first text-to-speech system for Walloon, an under-resourced Gallo-Romance language, using a VITS generative architecture and recordings from two speakers.
- Designed controlled experiments comparing grapheme- and phoneme-based models, Walloon-only training, and fine-tuning from French-pretrained models under full- and limited-data conditions.
- Found that transfer learning helped most in the 18-minute low-data condition but offered little advantage with approximately 156 minutes of Walloon speech; phoneme input gave additional pronunciation control without a clear overall quality gain.
- Evaluated the systems with complementary objective speech metrics and a perceptual study involving 23 native Walloon speakers; the best system achieved a 4.22/5 mean opinion score.
- Published the work as first author at SSW 2025 and released the models, code, and demonstrations publicly.

Synthetic French NER for Clinical Trial Texts

Feb. 2024

Final project, Hands-On NLP, Université Paris-Saclay M1 AI

- Co-developed a cross-lingual pipeline that translated English clinical-trial criteria, fine-tuned XLM-RoBERTa for French named-entity recognition, evaluated multilingual transfer, and released the resulting dataset on Hugging Face. Repository

Mobile Robotics Laboratory, University of São Paulo, Brazil

Mar.–Nov. 2014

Robotics Research Assistant during academic exchange

Supervisors: Marcelo Becker (laboratory head) and John Faber Archila-Diaz (project supervision)

- Contributed to a proof-of-concept semi-autonomous agricultural rover developed with the Brazilian Agricultural Research Corporation (EMBRAPA) to support LIBS-based soil-composition measurements and transmit field data for precision-fertilization decisions.
- Designed and integrated the rover's propulsion and steering control system: selected actuators, motor controllers, and sensing equipment under power and environmental constraints; implemented multidirectional movement and collision-prevention routines; and iterated through prototype and controlled field testing.

Publications

Espinosa Orjuela, J. F., Boula de Mareüil, P., and Evrard, M. (2025). "Speech synthesis for Walloon, an under-resourced minority language." *13th Speech Synthesis Workshop (SSW 2025)*, 189–195. *Peer-reviewed conference paper*. doi:10.21437/SSW.2025-29 | Code and models

Espinosa Orjuela, J. F., Pinto, R. M., Saavedra Guerra, J. L., and Becker, M. (2014). "Sistema de acionamento do freio de um carro autônomo." *22nd SIICUSP, University of São Paulo. Undergraduate symposium proceeding*.

Official USP record

Saavedra Guerra, J. L., **Espinosa Orjuela, J. F.**, Rueda Sánchez, O. E., and Becker, M. (2014). "Controle da propulsão e o estereçamento de um Rover aplicado à agricultura de precisão." *22nd SIICUSP, University of São Paulo. Undergraduate symposium proceeding*.

Official USP record

Manuscripts

Espinosa Orjuela, J. F., Payares, E., Puertas, E., Martinez-Santos, J. C., and Boula de Mareüil, P. (2026). “CLSE: Cross-Linguistic Evidence from Sublexical Embeddings.” *Manuscript under review*.
Contribution: First-author investigation of the structure preserved in phoneme-, syllable-, and word-level embedding spaces across Spanish, English, and French using probing, clustering, and representational-similarity methods.

Payares, E., **Espinosa Orjuela, J. F.**, Puertas, E., and Martinez-Santos, J. C. (2026). “PETER: Phonestheme Encoder from Transformer Embedding Representation.” *Manuscript in preparation*.
Contribution: Developing a multilingual Transformer encoder that combines phonestheme tokens with word-level semantic and positional information; final experiments and results validation are in progress.

Selected Presentations

Oral presentation: “Speech synthesis for Walloon, an under-resourced minority language.” *13th Speech Synthesis Workshop (SSW 2025)*, Leeuwarden, Netherlands, Aug. 2025.

Poster presentation: “Text-to-speech synthesis in Walloon, an endogenous language of Belgium.” *Language Technologies for All (LT4All 2025)*, UNESCO Headquarters, Paris, France, Feb. 2025.

Professional Experience

AI Evaluation & Data Contractor, Remote Nov. 2025–Jun. 2026
Independent Contractor

- Evaluated LLM outputs for RLHF and RLVF through pairwise comparisons, rubric-based scoring, written rationales, and quality-assurance audits using sampling and inter-annotator calibration.
- Analysed recurring failures in factual accuracy, reasoning, evidence use, safety, and instruction following.

Capital Fund Management (CFM), Paris, France Mar. 2025–Sep. 2025
AI Engineer Intern, LLM Systems and Decision Support – End-of-Studies Internship

- Developed and evaluated an analyst-facing conversational system that translated natural-language research questions into executable SQL for financial-data analysis, achieving 87.5% end-to-end accuracy on a 48-question internal benchmark.
- Designed the agent to determine which information and tools were needed before SQL generation, progressively narrowing the database context instead of exposing the generator to the full schema metadata.
- Implemented hybrid retrieval combining BM25, dense embeddings, and Reciprocal Rank Fusion to identify relevant tables in complex database documentation, reaching 95% top-k accuracy.
- Used smaller and lower-cost language models where benchmark performance remained comparable, while improving retrieval, context construction, and prompting; used benchmark results and LangSmith traces to analyse failures and monitor latency and cost.

Teaching & Mentoring

Mathematics and Physics Tutor – PAMRA, Universidad Industrial de Santander During B.Sc.

- Delivered approximately 400 hours of individual and group tutoring for university students in mathematics and physics, including sessions with more than 50 students.
- Prepared exercises and class materials aligned with university course programs in coordination with other tutors and the PAMRA program coordinator.

Technical Skills & Languages

Programming	Python, C/C++, SQL, Bash, TypeScript, JavaScript.
Machine Learning	PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Hugging Face Transformers.
AI Systems	LangChain, LangGraph, LangSmith, Ollama.
Databases	PostgreSQL, pgvector, Oracle, Redis, ChromaDB.
Computing	Linux, Git, Docker, Jupyter, Slurm.
Languages	Spanish (Native), English (C1), French (B2), Portuguese (B2).

Earlier Professional Experience

Self-Employed Contractor, Sydney, Australia 2018–2023
Managed residential construction and renovation projects.
Official records and publications use the full name José Felipe Espinosa Orjuela.